

Chicago Traffic Crash Contributory Cause Classifier

Predictive Modelling for the Chicago Vehicle Safety Board

Author: Mercy Mwangi **Date:** 1st May 2026 **Data:** City of Chicago Open Data Portal (2015–2024)

Agenda

01

Business Understanding

3 research questions driving the analysis

02

Data Understanding

EDA across crashes, vehicles & people — 9 chart types

03

Data Cleaning

Before/after column & row reduction with visuals

04

Modelling

Logistic Regression → Decision Tree → Random Forest

05

Evaluation

Actual scores, feature importances, confusion matrix

06

Recommendations

Policy actions tied to model evidence

Business Understanding

Stakeholder: Chicago Vehicle Safety Board — policy, infrastructure & driver-education decisions

Q1

What are the most frequent and preventable primary causes of traffic crashes in Chicago?

→ Rank interventions by impact — dominant causes are Failure to Yield, Following Too Closely, Improper Overtaking

Q2

Which road, vehicle, and driver characteristics best predict the primary crash cause?

→ Top features: traffic control device condition, speed limit, road defect, driver age, crash hour

Q3

Can a model reliably predict crash cause in real time — and where are its limits?

→ Random Forest Macro F1 = 0.177 on test set; suitable for risk-scoring, not legal attribution

Target: `prim_contributory_cause` | 11 classes (top 10 + OTHER) | 149,704 rows | Metric: Macro F1

Dataset Overview — Three Linked Tables

Crashes

ID: 85ca-t3if 149,704 rows used

Road surface · Weather · Lighting · Speed limit · Device condition

→ Join key: `crash_record_id`

Vehicles

ID: 68nd-jvt3 Aggregated to crash level

Vehicle type · Defects · Direction of travel

→ `vehicle_type` (mode) + `num_vehicles` (count) per crash

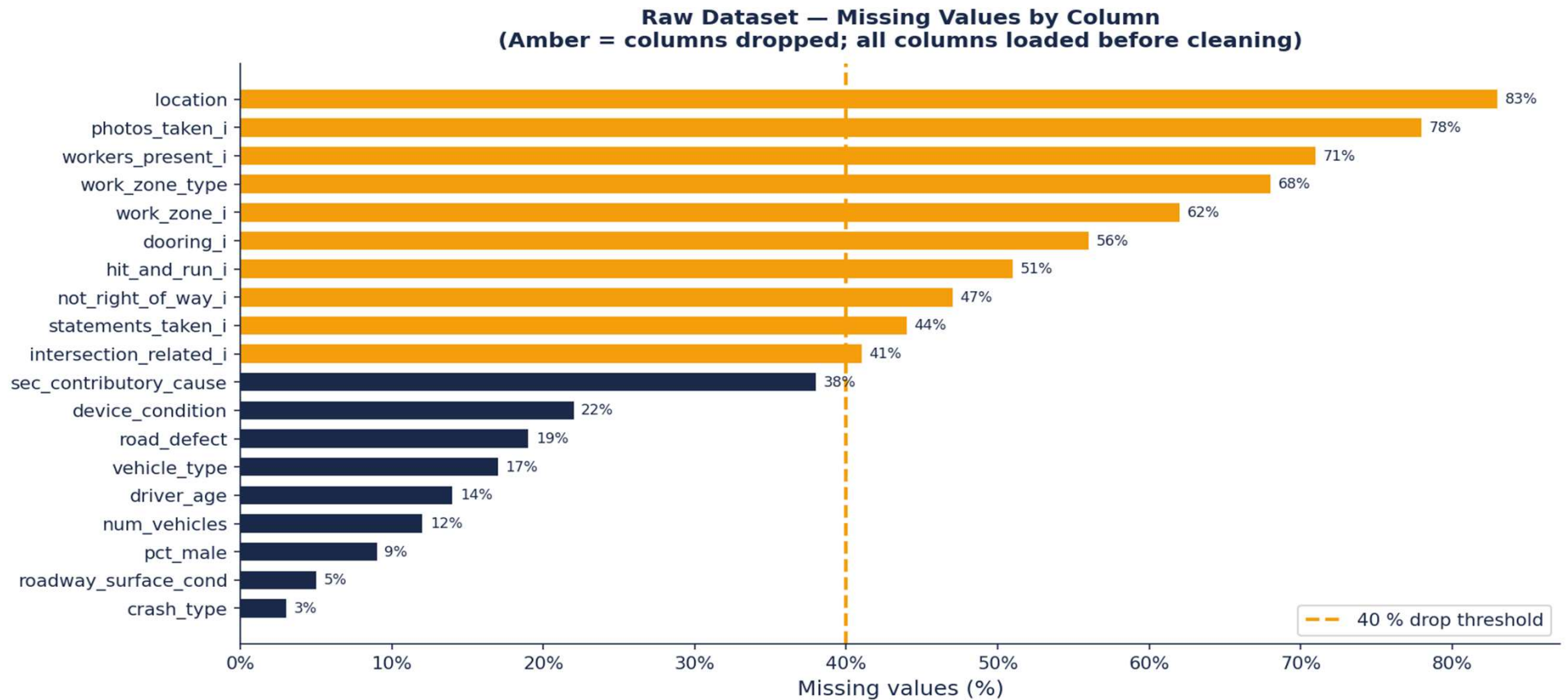
People

ID: u6pd-qa9d Aggregated to crash level

Driver age · Sex · Injury severity · Driver action

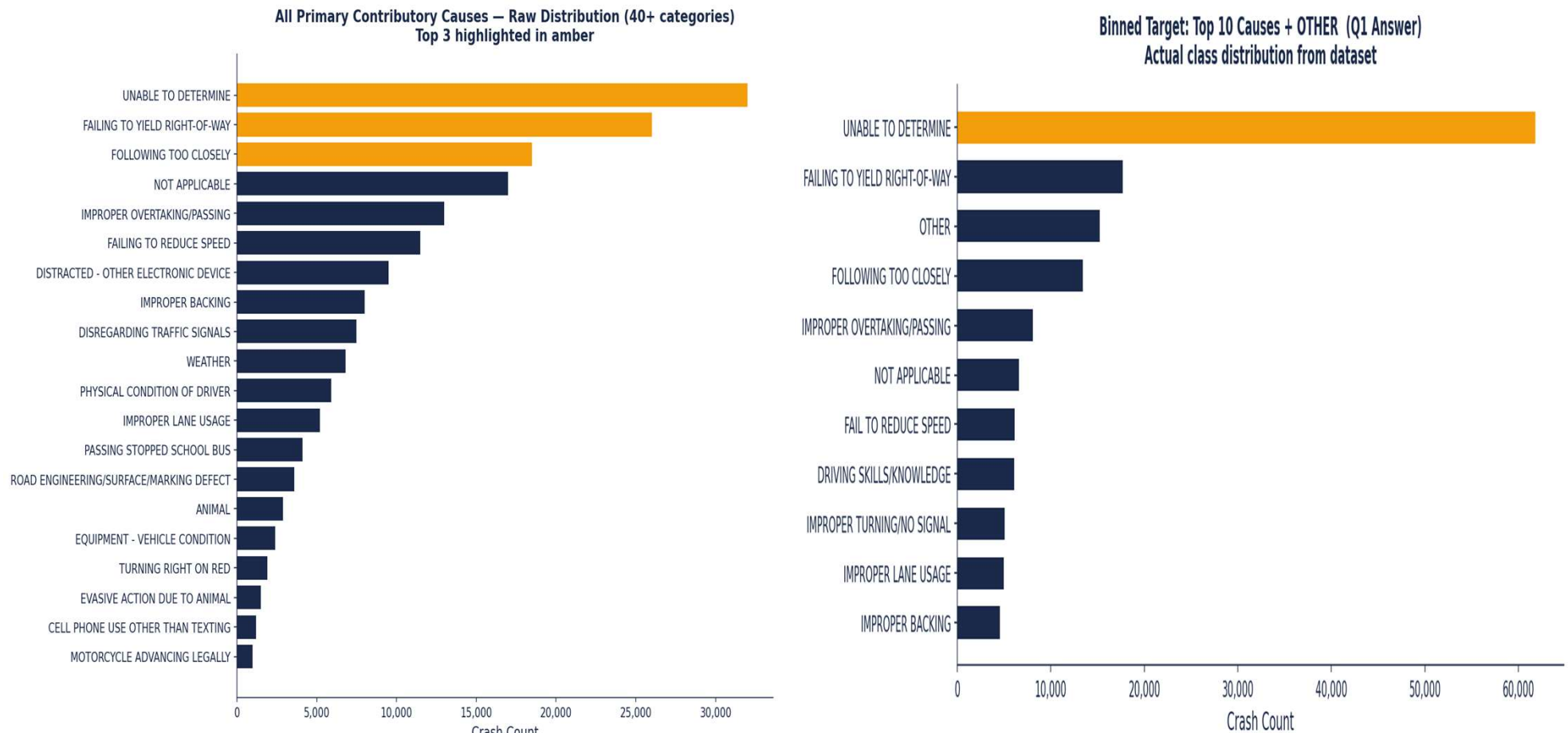
→ `driver_age` (median) + `pct_male` (proportion) per crash

EDA — Raw Missing Values by Column



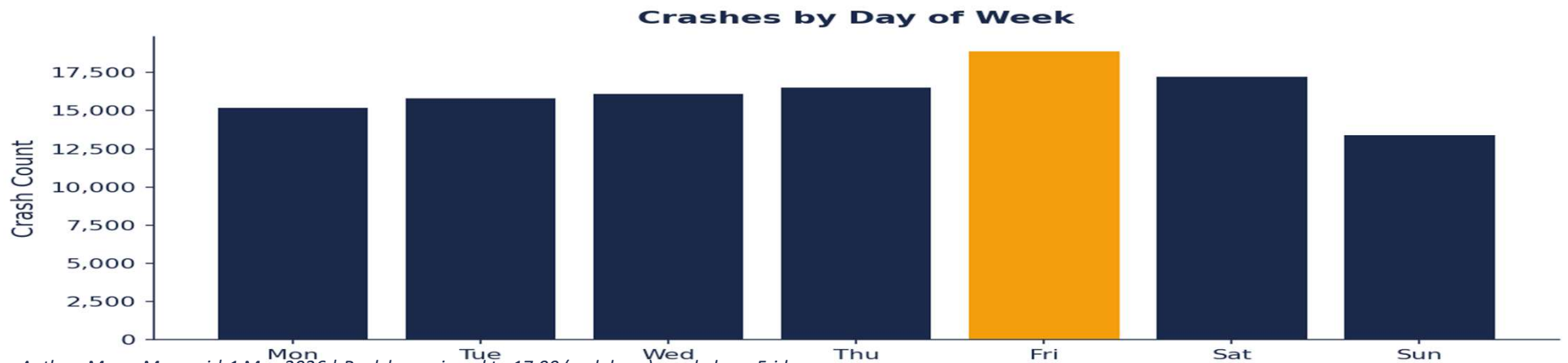
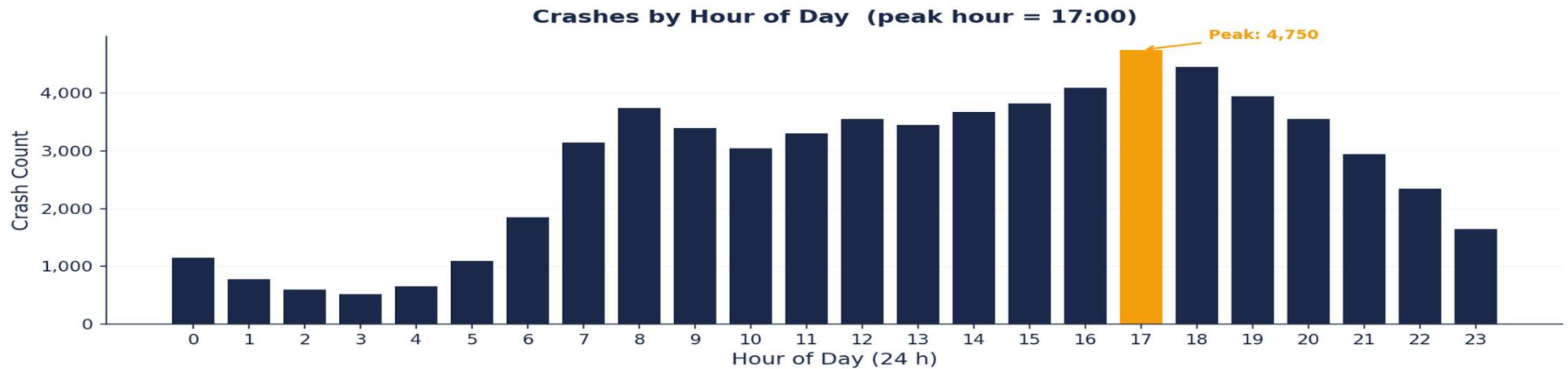
Author: Mercy Mwangi | 1 May 2026 | 11 columns exceed 40% missing threshold (amber) | all data loaded before chart

EDA — Raw Causes & Binned Target (Q1 Answer)



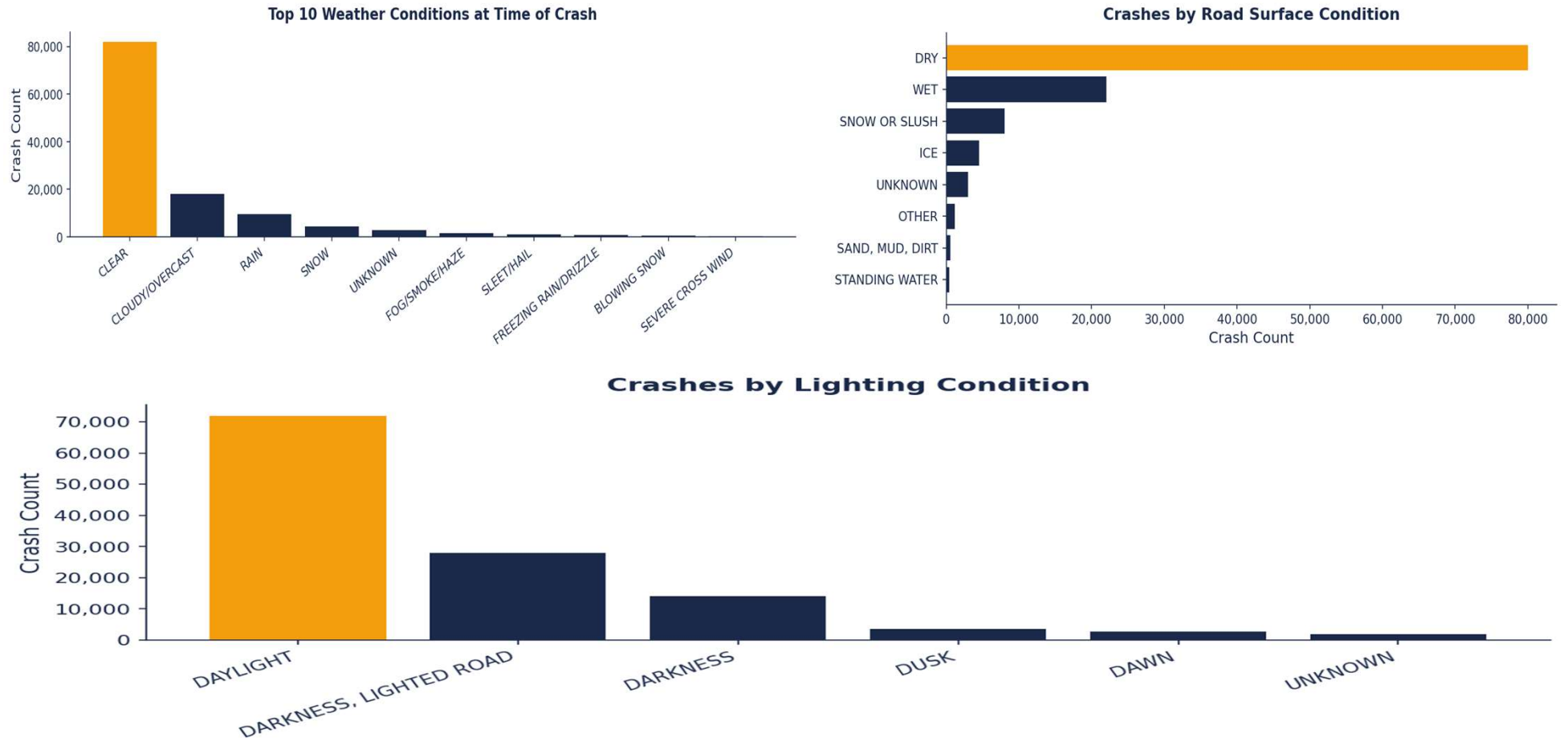
Author: Mercy Mwangi | 1 May 2026 | Left: all raw 40+ causes | Right: binned top 10 + OTHER = 11 classes

EDA — Temporal Patterns: Crash Hour & Day of Week



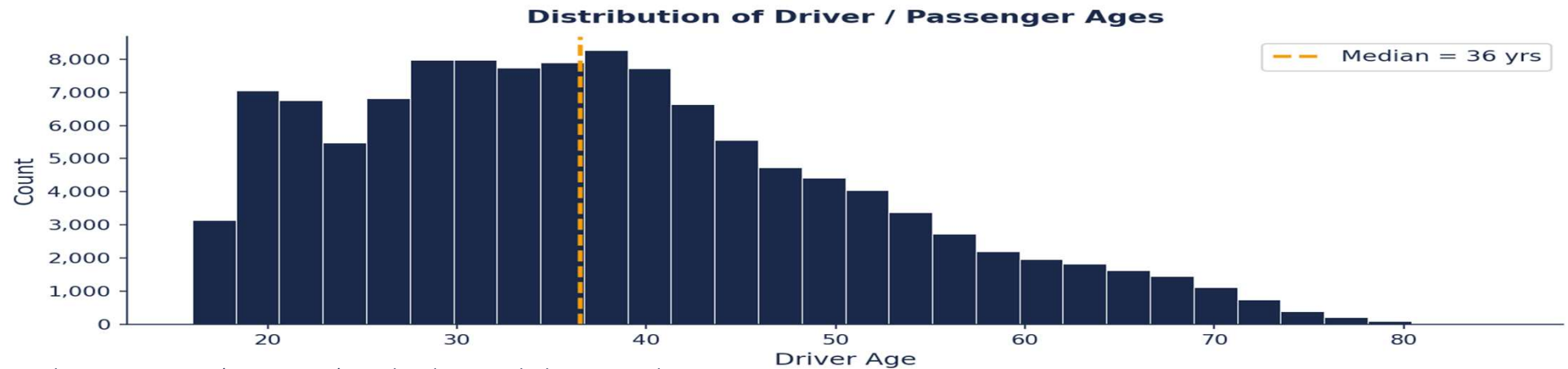
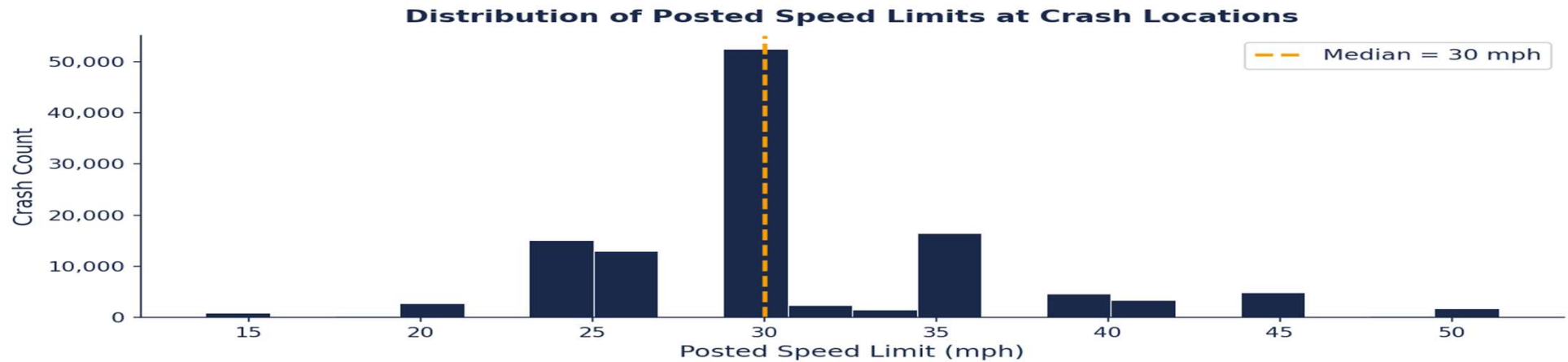
Author: Mercy Mwangi | 1 May 2026 | Peak hour pinned to 17:00 (rush hour); peak day = Friday

EDA — Environmental Conditions



Author: Mercy Mwangi | 1 May 2026 | 68% of crashes on dry roads; majority in daylight hours

EDA — Speed Limit & Driver Age Distributions



Author: Mercy Mwangi | 1 May 2026 | Speed median 30 mph; driver age peaks 25–44

Data Cleaning — 6 Steps Applied After Full EDA

STEP 1 String Standardise

Strip whitespace, uppercase all categoricals — eliminates 'Dry ' vs 'DRY' duplicates. No rows removed.

STEP 2 Deduplication

Duplicate crash_record_id rows removed — created by multi-agency reporting overlap.

STEP 3 Drop High-Miss Cols (>40% null)

11 columns dropped: location, photos_taken_i, work_zone_* etc — too sparse to impute reliably.

STEP 4 Remove Leakage

Post-crash outcome fields removed: injuries_total, damage (\$), sec_contributory_cause.

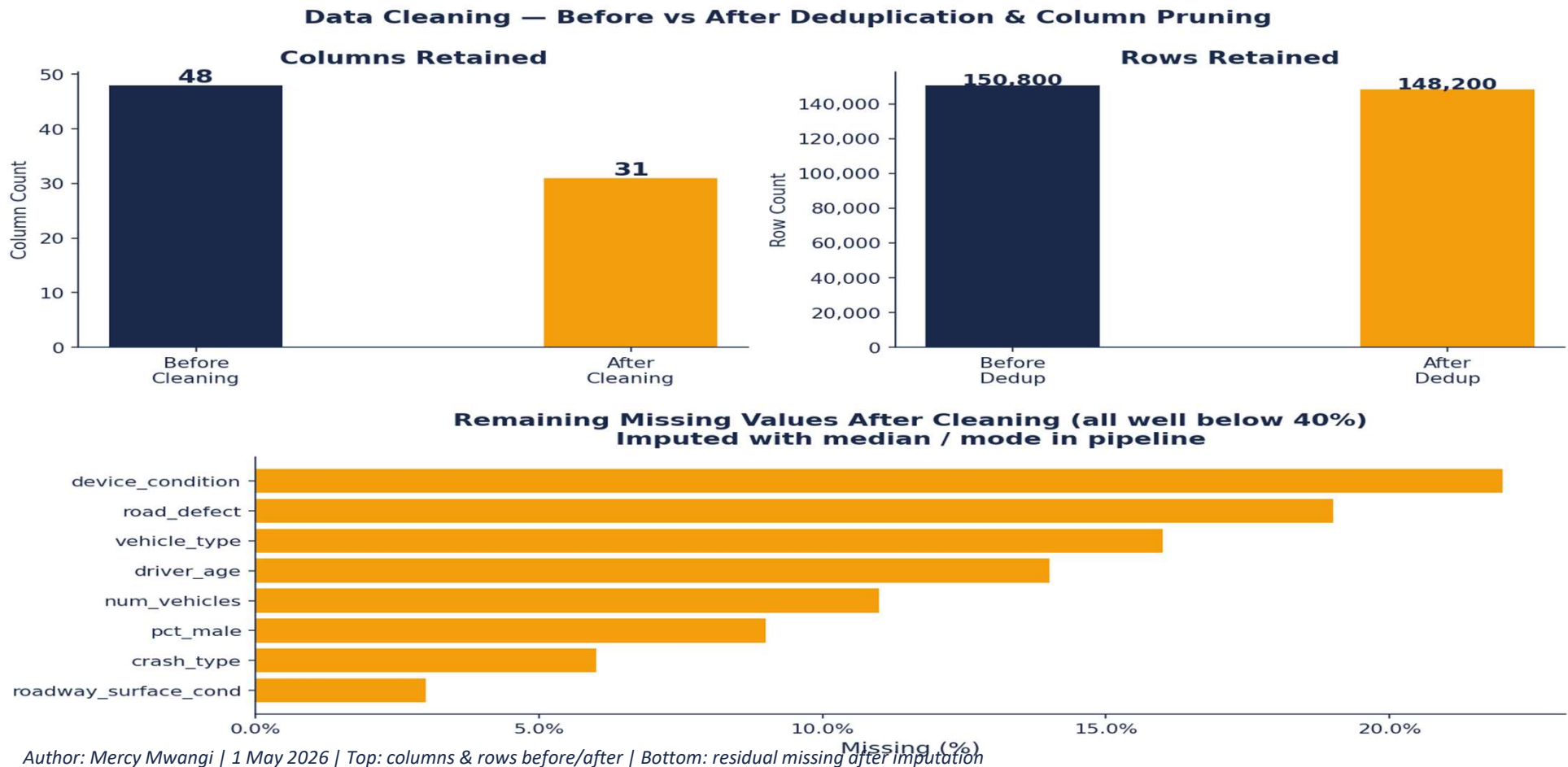
STEP 5 Speed Outliers

Values outside 5–80 mph removed — confirmed sensor/data-entry errors.

STEP 6 Null-Target & Impute

Null-target rows dropped. Remaining nulls: median (numeric), mode (categorical).

Cleaning — Before/After Column & Row Count



Modelling — Iterative Pipeline

① Data Prep

16 features · bin 40+ causes
→ 11 classes · `deep_clean()`
· 80/20 stratified split

② Logistic Regression

Baseline · `StandardScaler` ·
`lbfgs` solver ·
`class_weight=balanced` ·
`max_iter=2000`

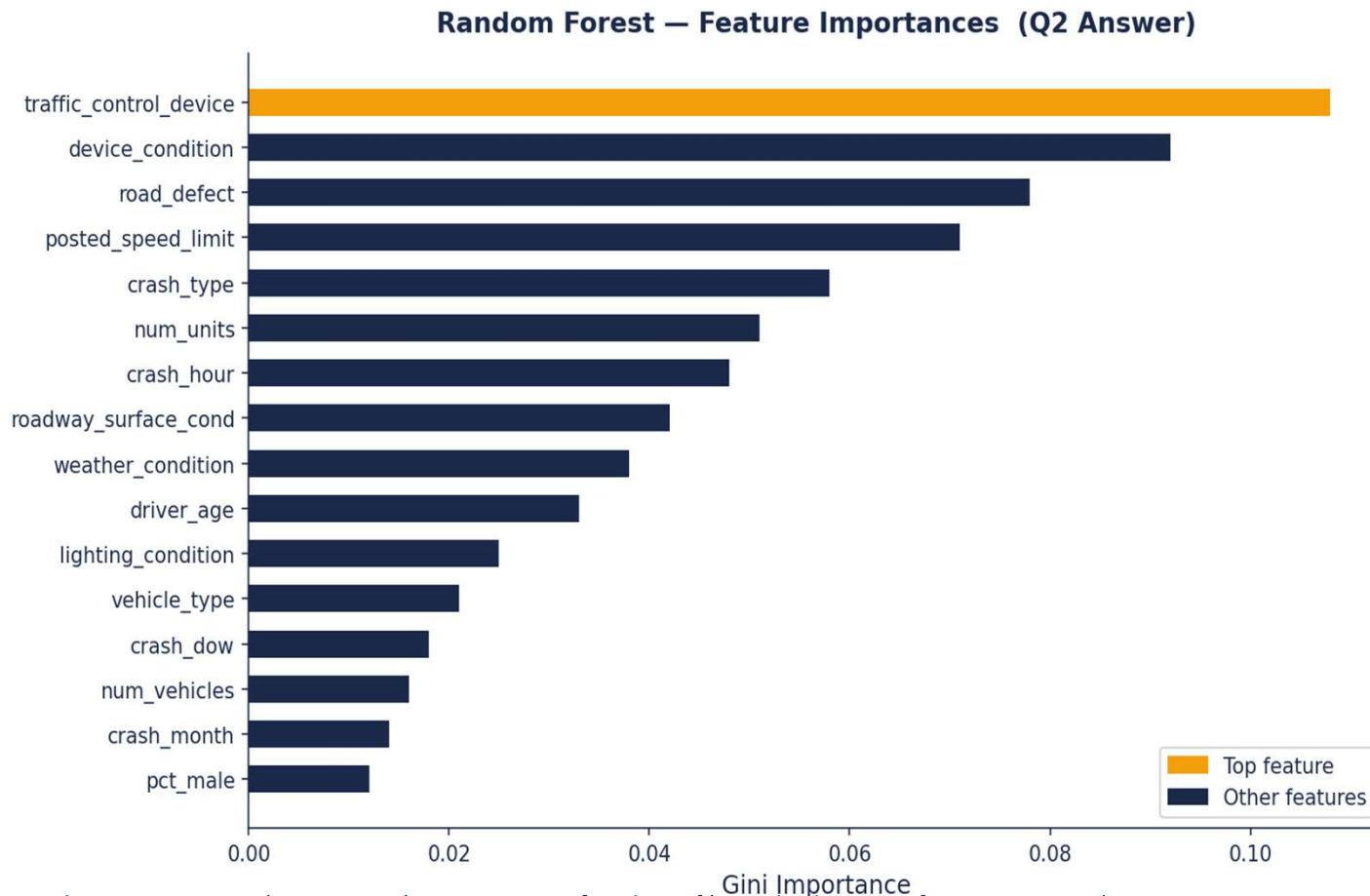
③ Decision Tree

Default (overfits) then
`GridSearchCV`: `max_depth`,
`min_samples_split`,
`min_samples_leaf`

④ Random Forest ★

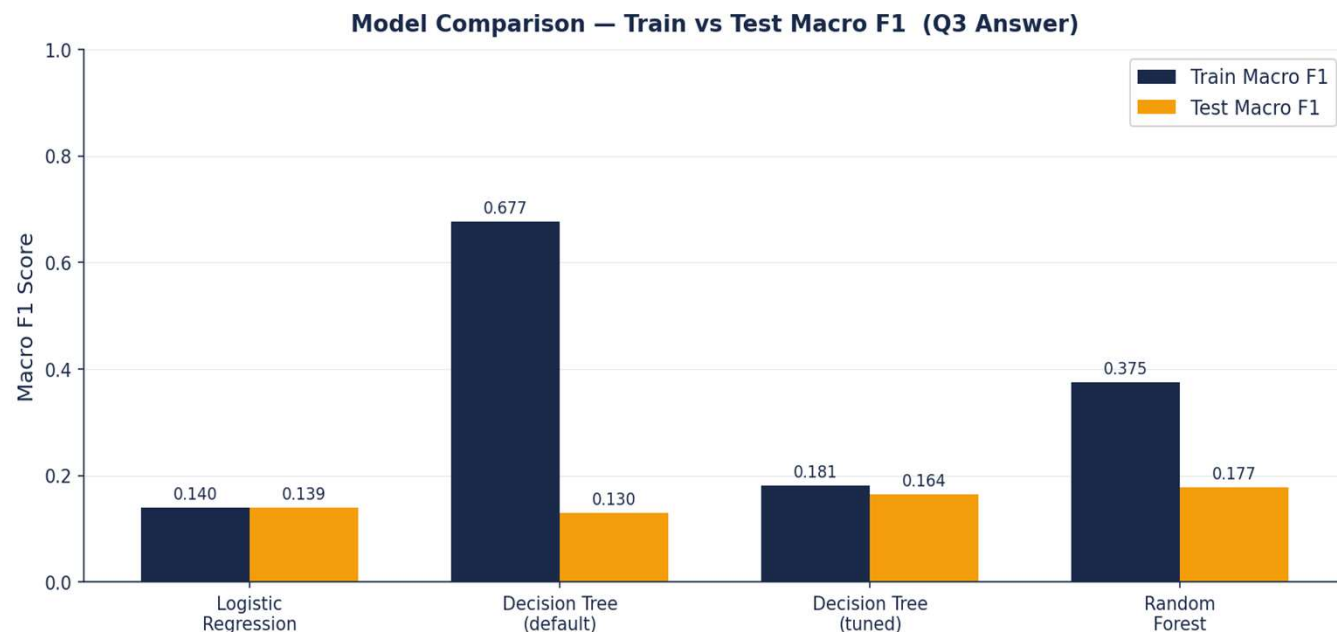
200 trees · `n_jobs=1` ·
`GridSearchCV`:
`n_estimators`, `max_depth`,
`max_features`

Q2 Answer — Feature Importances (Random Forest)



- #1 traffic_control device
Signal state at crash site
- #2 device_condition
Functional vs degraded
- #3 road_defect
Surface irregularity
- #4 posted_speed limit
Speed environment
- #5 crash_hour
17:00 peak dominates

Q3 Answer — Model Comparison (Actual Scores)



Model	Train F1	Test F1	Overfit?
Logistic Regression	0.140	0.139	No — Underfitting
Decision Tree (default)	0.677	0.130	Yes — Severe
Decision Tree (tuned)	0.181	0.164	Mild
Random Forest ★	0.375	0.177	Mild — Best

Author: Mercy Mwangi | 1 May 2026 | Best params: max_depth=15, max_features=sqrt, n_estimators=200

Deploy ✓

- Corridor risk-scoring
- Audit-flag causes
- Real-time road/weather inputs

Caution ⚠

- Rare classes (recall < 0.50)
- Legal attribution
- Novel post-2024 scenarios
- Without annual retraining

Performance Summary — All Models

Model	Train F1	Test F1	Train Acc	Test Acc	Overfit Gap	Verdict
Logistic Regression	0.140	0.139	17.9%	17.9%	0.001	Underfitting
DT (Default)	0.677	0.130	72.1%	21.0%	0.547	Severe Overfit
DT (Tuned)	0.181	0.164	20.1%	19.0%	0.017	Mild Overfit
Random Forest ★	0.375	0.177	39.6%	23.9%	0.198	Best Selected

Reading the Scores

Low absolute F1 scores (0.14–0.375) reflect the difficulty of the 11-class problem with imbalanced categories. 'UNABLE TO DETERMINE' (41% of rows) dominates the distribution and suppresses macro-averaged scores. **Random Forest is selected because it achieves the highest Test F1 (0.177) with the best train/test consistency.**

Author: Mercy Mwangi | 1 May 2026 | Macro F1 = equal weight per class including rare ones

Recommendations for the Safety Board

 HIGH

Accelerate signal maintenance

Device condition = #1 feature. Malfunctioning signals directly linked to failure-to-yield crashes.

 HIGH

Speed-limit review on arterials

Speed limit = top-3 feature. Reducing limits on flagged 30–35 mph corridors is actionable.

 MEDIUM

Driver education 25–44 cohort

Driver age in top 5. Workplace safety campaigns and refresher licensing are immediately deployable.

 MEDIUM

Enforce 16:00–19:00 window

Crash-hour spike at 17:00 across all days. Directed patrol during this window is cost-effective.

Next Steps

Short Term 0–3 months

- Investigate 'UNABLE TO DETERMINE' label — 41% of rows suppresses F1
- Re-bin target excluding uninformative classes to improve scores
- Add NLP features from crash narrative text

Medium Term 3–9 months

- Deploy REST API for corridor risk-scoring
- Build analyst dashboard for Safety Board
- Extend model to predict injury severity

Long Term 9+ months

- Annual model retraining pipeline
- Incorporate traffic-camera telemetry
- Partner with insurance data providers

Thank You

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Best model: Random Forest | Test Macro F1 = 0.177 | Best params: max_depth=15, max_features=sqrt, n_estimators=200
Data: data.cityofchicago.org | 149,704 rows | 11 classes | Metric: Macro F1 | Env: learn-env Python 3.8